

Did Employer Skill Language Shift After the Generative AI Inflection Point? *U.S. Federal Job Postings (2019–2026)*

Research Project Proposal

Md Khalid Bin Kamal
Tanmoy Biswas
Tryphosa Htoi Sam Roi
Musharraf Mir Akhmadova

EPPS 6323 – Knowledge Mining
March 3, 2026

Research Question

Main Question

Did employer skill language shift after the diffusion of generative AI in late 2022?

Sub-questions

- Did AI-related terminology increase?
- Did technical skill intensity change?
- Did emphasis on soft / coordination skills change?
- Did the overall topic structure of job descriptions shift?

Motivation

- Generative AI diffused rapidly after November 2022.
- Public debate focuses on employment loss and wages.
- Much less attention to how employer expectations evolve.
- Job postings provide real-time signals of labor demand.
- If technology restructures tasks, we should observe language shifts in job descriptions.

Theory

- Technological change restructures tasks, not just employment (Autor et al., 2003).
- Automation substitutes routine work but increases demand for complementary skills (Acemoglu & Restrepo, 2020).
- Technology-intensive production raises demand for cognitive and coordination skills (Deming, 2017).
- If generative AI automates routine cognitive tasks, employer skill language should shift after late 2022:
 - ↑ AI task language
 - ↑ Complementary skills
 - ↓ Routine cognitive tasks

Research Design

Identification Strategy

- Generative AI diffusion represents a common technological shock.
- Occupations differ in their pre-existing exposure to AI capabilities.
- Late 2022 marks a visible acceleration in public adoption and organizational awareness of generative AI tools such as release of ChatGPT.

Exposure Measure: AI Occupational Exposure Index (AIOE)

- Developed by Felten, Raj & Seamans (2021)
- Using O*NET data links AI applications to occupational abilities
- Occupation-level
- Time-invariant
- Continuous
- Independent of post-2022 outcomes

Research Design

Empirical Design

- Common shock: Generative AI diffusion
- Treatment intensity: AIOE (continuous, occupation-level)
- Unit of analysis: Occupation $\times$ Month
- Method: Continuous Difference-in-Differences
- Shock timing : Nov 2022
- Pre: Jan 2019 – Oct 2022
- Post: Nov 2022 – 2026

Research Design

Baseline DiD model

$$Y_{it} = \alpha + \beta (Post_t \times AIOE_i) + \gamma_i + \delta_t + \epsilon_{it}$$

Where:

Y_{it} = skill intensity or topic prevalence

$Post_t$ = 1 after November 2022

$AIOE_i$ = occupation-level AI exposure

γ_i = occupation fixed effects

δ_t = time fixed effects

β measures whether occupations with higher AI exposure experienced larger post-2022 shifts in employer skill language.

Exposure is continuous to preserve cross-occupational variations.

Research Design

Event study (Parallel Trends & Dynamics)

$$Y_{it} = \alpha + \sum_{\{k \neq -1\}} \beta_k (D_t^k \times AIOE_i) + \gamma_i + \delta_t + \varepsilon_{it}$$

Rationale:

- Test for differential pre-trends
- Examine timing of language shifts
- Observe whether restructuring is immediate or gradual
- This allows us to assess whether divergence emerges only after late 2022.
- Omitted period: the month before Nov 2022.

Research Design

Knowledge Mining Techniques

Text Processing

- Cleaning
- Tokenization
- Lemmatization

Feature Construction

- AI-specific keyword dictionary
- Technical skill dictionary
- Soft skill dictionary
- TF-IDF weighting

Advanced Analysis

- Topic Modeling (LDA)
- Topic prevalence comparison (Pre vs. Post)

Data

Primary Source: U.S. Federal Job Postings (2019–2026)

Variables available:

- Posting date
- Occupation codes
- Salary range
- Full job description text

Balanced pre- and post-AI periods allow temporal comparison.

**We are exploring additional data sources, such as Muse (2024-2026), for external validation.*

Robustness Checks

- Parallel Trends: Event study to test for differential pre-trends
- Alternative Timing: Test lag structures and post definitions
- Placebo Tests: Assign pre-2022 shock dates
- Alternative Specifications: Different aggregation levels; exclude extreme exposure
- Inference: Cluster standard errors at occupation level

Expected Contributions & Challenges

Expected Contributions

- Provides early evidence on how generative AI reshapes employer expectations.
- Moves beyond employment counts to task and skill restructuring.
- Combines causal design with computational text analysis.

Challenges (So Far)

- Data related: Such as defining treatment timing (diffusion vs. release date), missing data and other mismatches.
- Ensuring parallel trends assumption.
- Measuring skill intensity consistently over time.
- Distinguishing “*AI hype language*” from substantive restructuring.

Thank You!

We welcome your questions and suggestions.